Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/13/2006
PI Name:	Baker, James R M.D.		
Project Title:	Biosensors for real-time monitoring of radiation induced biologic effects in space.		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-0648	**Congressional District:**	15
Comments:	The Center for Biologic Nanotechnology applies nanotechnology to solving biological problems of medical interest.		
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:	Baker, James R.		
COI Name (Institution):			
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	We propose to develop nanoscale biosensors and bioactuators for use in astronaut health and safety monitoring, in conjunction with external analytic bioNEMS devices. This would involve the evolution of nanoscale polymer structures less than 20 nm in diameter as the basis of the sensor/actuators. The structures would be designed to target into specific cells of an astronaut and be able to monitor health issues such as the exposure to radiation or infectious agents. These molecules would also be able to administer therapeutics in response to the needs of the astronaut, and act as actuators to remotely manipulate an astronaut as necessary to ensure their safety. An example of this latter capability would be to activate muscle movement causing an unconscious astronaut to walk back into a spacecraft and out of harm's way. In addition, the nanosensor could be used to develop cellular-based biochips that would be useful as sensors for the remote detection of life or other scientific analysis in space. These studies will be truly multidisciplinary, involving a fusion of disciplines including nanotechnology-based materials science, bioengineering, bioinformatics and medical sciences. We will use these different disciplines to converge on the		

Task Description:

design and manipulation of the nanosensors, and the development of non-invasive systems to interact with the sensors through functional MRI or multispectral fluorescence imaging. This latter task would require the engineering of wearable NEMS systems by engineers involved in this project while other members of the project team would test the efficacy of these systems using biologic models, such as experimental animals. Because of these broad requirements, the research would involve the multidisciplinary team from the Medical, Engineering and basic science (LS&A) schools at the University of Michigan, and would train multidisciplinary scientists at the pre-and postgraduate level. We will partner with a historically black institution, Howard University, for supplemental materials science capability necessary for this program. This team would develop strong interactions with NASA scientists such as the Nanotechnology group at the Ames research center and multispectral imaging scientists, and allow NASA scientists opportunities for training at the lead institution. Objective I: The development of dendritic polymer based biosensors capable of monitoring radiation-induced changes in lymphocytic populations. Objective II: The analysis and verification of the radiation sensing capability in lymphocytes in vitro. Objective III: The development of a noninvasive laser analysis system to monitor biosensor signals of radiation-induced damage in lymphocytes. Exposure to ionizing radiation is an important problem in manned space travel. Exposure to radiation can cause both acute and chronic illness, and the spacecraft itself provides little shielding to the radiation that could be encountered in space. Acute, high-level radiation exposure of greater than 2500 cGy will inactivate lymphocytes and induce apoptosis in the majority of these cells, leaving the individual susceptible to infectious diseases (American Journal of Industrial Medicine 20:805-10, 1991, Stem Cells 2:275-85, 1997). Higher doses of radiation can prevent the replication of cells in the respiratory and gastrointestinal mucosa as well as platelets and granulocytes (Military Medicine 165:275-7, Progress in Clinical & Biological Research. 372:155-65, 1991, British Journal of Hospital Medicine 39:498-504, 1988). This leaves the subject extremely ill and death usually occurs due to bone marrow failure or hemorrhage from a combination of inadequate platelet function and mucosal breakdown. Low-level radiation exposure can also cause illness, in a more protracted manner with serious heath compromise. This type of radiation exposure can cause genetic damage that can predispose to a number of neoplasms, particularly lymphomas (Health Physics 56:821-38, 1989, Medicine, Conflict & Survival 14:106-19, 1998). Lymphocytes are sensitive to radiation effects and therefore are likely to have radiation-induced damage that can cause neoplasms (Eisenberg RL. Radiation injury and protection. In: Radiology: an illustrated history. St. Louis: Mosby Year Book; p. 156-82, 1992). However, with prolonged exposure to low doses of radiation many types of cancers can be observed (Radiation Research 144:160-9, 1995).

While it is medically important to monitor real-time radiation exposure in individuals, no techniques are currently suitable to do this. There are radiation monitoring devices in spacecraft, but these do not measure actual dosing to the astronauts or anticipate the biologic consequences of exposure. In addition, it is difficult to calculate the radiation dose that an astronaut might accumulate in the multiple environments he or she encounters in space (either in or outside of the spacecraft). Individuals also may have different susceptibilities to radiation-induced illness, making the measure of a specific “biological dose” for each individual imperative (International Journal of Radiation Oncology, Biology and Physics 30:997-1002, 1994, International Journal of Radiation Oncology, Biology and Physics 42:379-89, 2001). There are, however, techniques to measure the “biologic dose” of ionizing radiation and these center around the sensitivity of lymphocytes to radiation exposure. This sensitivity allows acute radiation exposure to be identified by first identifying the percentage of lymphocytes undergoing apoptosis in peripheral blood and subsequently through monitoring lymphocyte counts for the gradual disappearance of the cells after undergoing cell death (Stem Cells 15:S269-74, 1997). Chronic radiation exposure at low-levels can also be evaluated in lymphocytes through the monitoring of cytogenetic changes (Radiation Research 140:134-42, 1994, Mutation Res 339:177-89, 1995, Radiation Research 146:479-82, 1996, Radiation Research 148:S33-8, 1997), although this is somewhat controversial (Radiation Research 147:506-10, 1997). A sensitive marker for chronic exposure is the frequency of reciprocal translocations in the chromosomes of blood lymphocytes which can be identified through fluorescence in situ hybridization (Radiation Research 33:S206-14, 1992). This is a consistent marker that is not affected by other metabolic or environmental toxins but requires that cells must be removed from the individual and analyzed either with microscopy or some type of analytic machine, such as a flow cytometer (Medical Physics 7:609-15, 1980). However, neither of these techniques can be performed in space because of the equipment required, the necessity for special processing of the cells plus the limitations of a zero gravity environment. Even if a device could be developed that would allow for the analysis of blood samples in zero gravity, the necessity to draw blood would still create technical difficulties. Importantly, this testing cannot be done continuously to provide real-time analysis of an individual’s radiation exposure in a manner that might allow for therapeutic intervention (such as transfusion of stored, shielded autologous blood or bone marrow) (N Engl J Med 321:205-12,1989).

Optimal Solutions for Radiation Monitoring in Space It would be optimal to monitor the biologic effects of radiation during space travel using a system that could measure exposure in a real-time, continuous manner. This should be done non-invasively as to avoid multiple sampling or blood drawing and therefore could provide momentary information on the astronaut’s status. By necessity, this monitoring approach would require some type of highly miniaturized device that could assess radiation effects in target cells, such as circulating lymphocytes, in a reliable and specific manner. It would also allow collecting data that would measure and differentiate acute as well as chronic exposure. A valuable addition would be that this system could measure the effects of other physiologic perimeters non-invasively. These applications might include calcium mobilization, as a measure of bone demineralization from zero gravity, or screening neutrophils for evidence of infection. This latter point is important since it became clear during the Mir mission that spacecraft do not have sterile environments, particularly after long-term inhabitation. Any system that would monitor these parameters non-invasively would also have to be able to operate in zero gravity space environment with low-power requirements and also with minimal weight and size to make it practical for inclusion on a long-term space voyage. These additional requirements emphasize the potential benefit of a monitoring system that could be adaptable to multiple monitoring applications avoiding the need for redundant technology.

We believe that the development of a “biologic flow cytometer” for the non-invasive monitoring of radiation-induced damage to lymphocytes would be an exciting and highly valuable goal to accomplish and could fulfill all the requirements of a non-invasive radiation monitoring system. We have developed similar types of nanosensors for other applications, documented that they insert and function within cells and can provide multi-spectral signals that can be measured on a real-time basis. Importantly, our colleagues have knowledge of a non-invasive laser system for monitoring that is small enough to even be wearable to provide continuous data from astronauts. Thus, based on preliminary work in other disease systems (see below), we have confidence that in vitro studies developing lymphocyte targeting biosensors and a demonstration of the concept of a laser-monitoring device can be accomplished within the three-year timeframe of the proposal.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This work seeks to develop cellular biosensors based on dendritic polymers. Nanoscale polymer structures less than 20 nm in diameter will be used as the basis of the biosensors. The structures will be designed to target into specific cells of an astronaut and be able to monitor health issues such as exposure to radiation. Multiple components can be assembled on the polymers including target directors, analytical devices (such as molecular probes), and reporting agents. The reporting will be accomplished through fluorescence signal monitoring with the use of multispectral analysis for signal interpretation. These nanosensors could facilitate the success and increase the safety of extended space flight. The design and assembly of these devices has been pioneered at the Center for Biologic Nanotechnology in the University of Michigan. We have developed a femtosecond laser-based optical system that allows both in vitro and in vivo fluorescence measurements of various biological samples, such as cells, viruses and single molecules. Using mouse models, we have conducted real-time noninvasive measurements of circulating metastasis breast cancer cells labeled with novel optical probes, such as multi-functional nanoparticles. Our work on this project involved selecting appropriate biosensing dyes that can be conjugated to targeting dendrimers; developing appropriate models for the testing of targeted, biosensing dendrimeric conjugates; and then establishing the physical characteristics, specificity, rate of uptake, and intracellular effects of those conjugates. Effective targeting of conjugates bearing biosensing dyes was established. These conjugates were confirmed to exhibit acceptable characteristics (photostability, specificity, rate of uptake, low toxicity) for use in in vitro models. Using the methods and model systems established, we determined the effect of gamma radiation on cultured human lymphocytes as monitored using the conjugated biosensors. Transmembrane potential fluorescence of irradiated cultured lymphocytes decreased significantly over 24 hours when exposed to moderate levels of radiation, indicating that the conjugates could signal this event. We have been developing a model for labeling cells with different fluorochromes for monitoring radiation-induced changes in lymphocytes in blood. Multiple and two-color dye systems were used throughout the study. The cell viability was assessed using the two-photon optical-fiber fluorescence (TPOFF) detection system. We have tested multiple cell types, including primary cells isolated from blood. Detection of apoptotic lymphocytes was accomplished with two-color dye. Additional one-component indicator and reporting dye for lymphocytes, SNARF-1, alleviated the necessity of balancing the fluorescence signal from two distinct dyes and allowed detection of non-apoptotic and apoptotic T-lymphocytes. We have developed an in vitro and in vivo surveillance of the expression of CD14 on the circulating cells. The increase of CD14-expressing cells in the circulation is an indicator of an inflammatory process. The model for staining cells with different dyes ex vivo was further developed to include cells stably expressing a fluorescent signal, thus eliminating possible leakage of a dye and a bystander staining effect. Due to their high fluorescent signal, these cells can be easily traced in vivo with the TPOFF. Dendrimer-based devices have been designed, synthesized, and characterized for sensing radiation-induced changes in lymphocyte populations. Additionally, these devices are excellent for studying events in vivo by using a two-photon fiber optic analytical procedure.
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